BASE CHANGE FORMULA FOR DERIVED SECTIONS OVER ETALE OPENS ALONG THE DERIVED DESCENT TOWER

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: homotopy colimit representation of sections by perfect complexes, etale invariance of sections over the level, and levelwise derived base change with homotopy colimit commutation

Editorial note. The label lemma-compute-over-affine-derived names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The representation of sections over a quasi-compact quasi-separated etale open as a homotopy colimit of sections against perfect complexes, valid for every object of the quasi-coherent derived category
- The derived base change lemma for derived sections against perfect objects along the derived descent tower
- Derived tensor with A_n commuting with homotopy colimits
- Invariance of sections under the etale base change identifying U_n over the level with U

Source reference. Original reference 0DID: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2753–2753. Prerequisite pointer invoked at proof line(s) 2847; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DIC: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2695–2695. Prerequisite pointer invoked at proof line(s) 2819; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DIB: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2668–2668. Prerequisite pointer invoked at proof line(s) 2818; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CRW: sources/stacks-chow/tags.txt, source lines 16011–16011. Prerequisite pointer invoked at proof line(s) 2830; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 0DIB let K be as in Lemma 0DIC. For any étale morphism $U \to X$ with U quasi-compact and quasi-separated we have*

$$R\Gamma(U, K) \otimes^{\mathbf{L}}_A A_n = R\Gamma(U_n, K_n)$$

in $D(A_n)$ where $U_n = U \times_X X_n$.

Proof. Fix n. By Derived Categories of Spaces, Lemma 0CRW there exists a system of perfect complexes E_m on X such that $R\Gamma(U, K) = \text{hocolim} R\Gamma(X, K \otimes^{\mathbf{L}} E_m)$. In fact, this formula holds not just for K but for every object of $D_{QCoh}(\mathcal{O}_X)$. Applying this to K_n we obtain

$$\begin{aligned} R\Gamma(U_n, K_n) &= R\Gamma(U, K_n) \\ &= \text{hocolim}_m R\Gamma(X, K_n \otimes^{\mathbf{L}} E_m) \\ &= \text{hocolim}_m R\Gamma(X_n, K_n \otimes^{\mathbf{L}} E_m|_{X_n}) \end{aligned}$$

Using Lemma 0DID and the fact that $- \otimes^{\mathbf{L}}_A A_n$ commutes with homotopy colimits we obtain the result. □